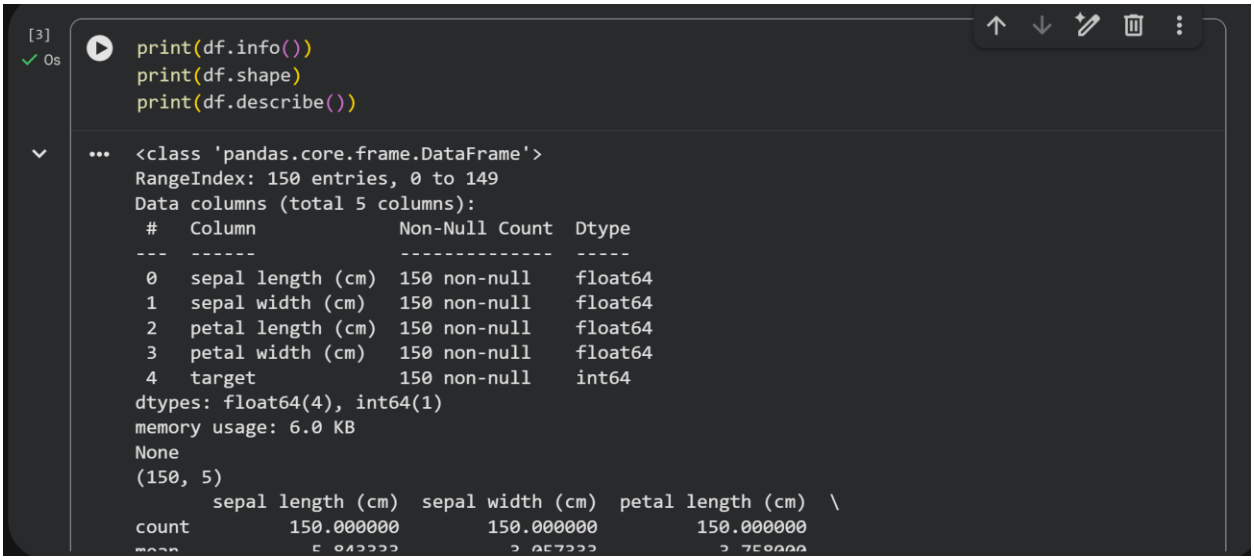
LAB # 04 DECISION TREE CLASSIFIER

Task 1:



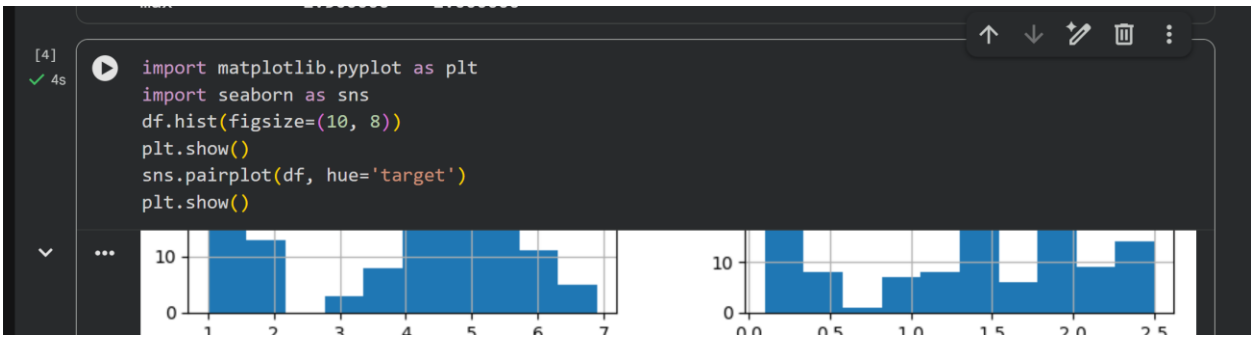
```
[1]
✓ 1s
from sklearn.datasets import load_iris
import pandas as pd
iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['target'] = iris.target
```

Task 2:

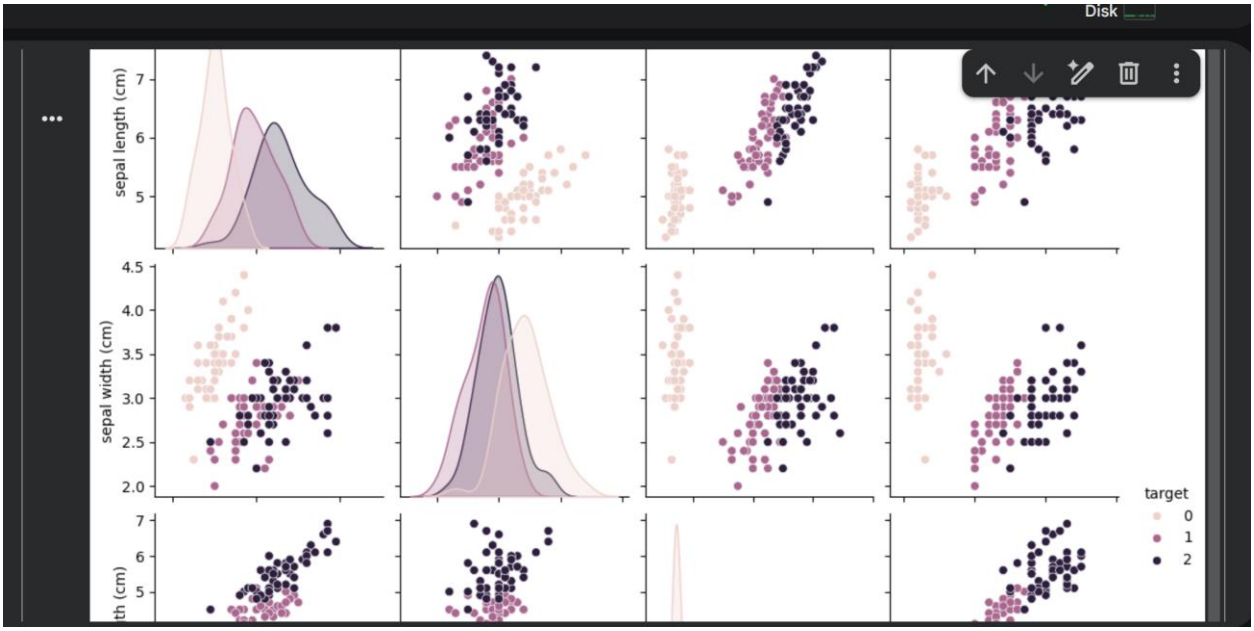


```
[3]
✓ 0s
print(df.info())
print(df.shape)
print(df.describe())

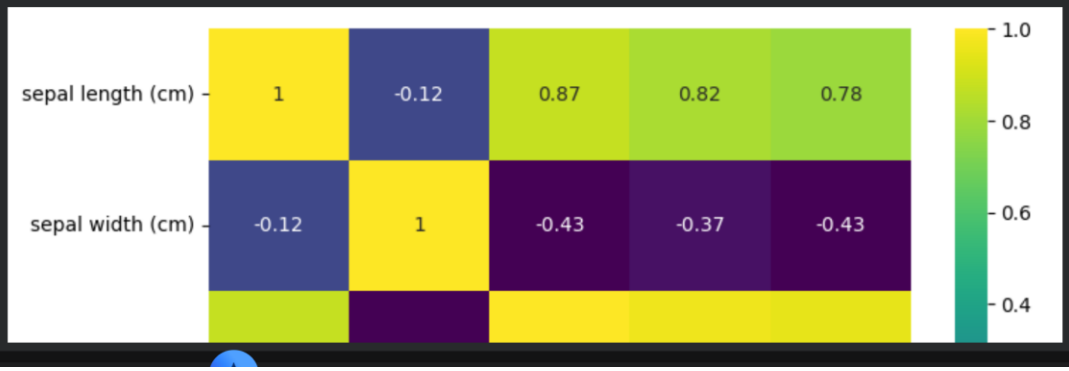
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   sepal length (cm)  150 non-null    float64
 1   sepal width (cm)   150 non-null    float64
 2   petal length (cm)  150 non-null    float64
 3   petal width (cm)   150 non-null    float64
 4   target             150 non-null    int64
dtypes: float64(4), int64(1)
memory usage: 6.0 KB
None
(150, 5)
       sepal length (cm)  sepal width (cm)  petal length (cm)  \
count         150.000000        150.000000         150.000000
```



```
[4]
✓ 4s
import matplotlib.pyplot as plt
import seaborn as sns
df.hist(figsize=(10, 8))
plt.show()
sns.pairplot(df, hue='target')
plt.show()
```




```
[5]
✓ 0s
plt.figure(figsize=(8, 6))
sns.heatmap(df.corr(), annot=True, cmap='viridis')
plt.show()
```



Task 3:

```
[6]
✓ 1s
from sklearn.model_selection import train_test_split
X = df.drop('target', axis=1)
y = df['target']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, random_state=42)
```

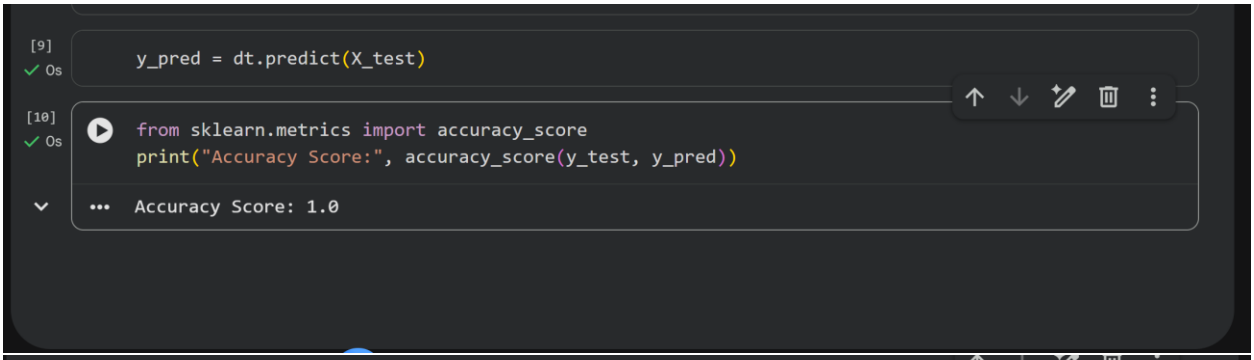
Task 4:



```
[7]
✓ 0s
from sklearn.tree import DecisionTreeClassifier
dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train, y_train)

DecisionTreeClassifier
DecisionTreeClassifier(random_state=42)
```

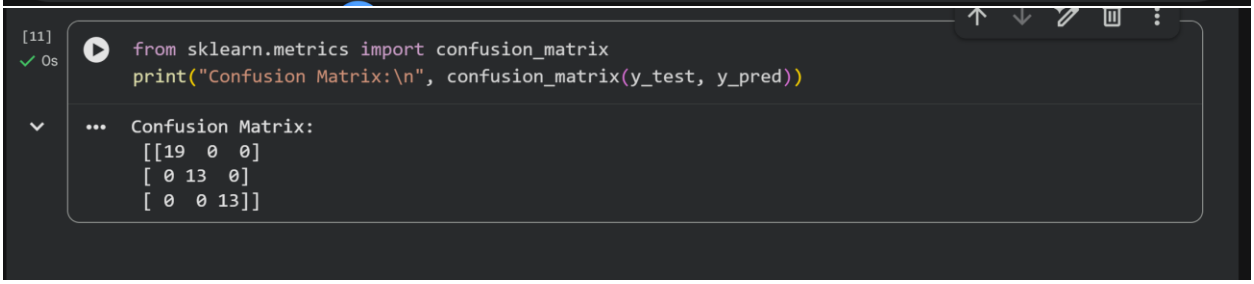
Task 5:



```
[9]
✓ 0s
y_pred = dt.predict(X_test)

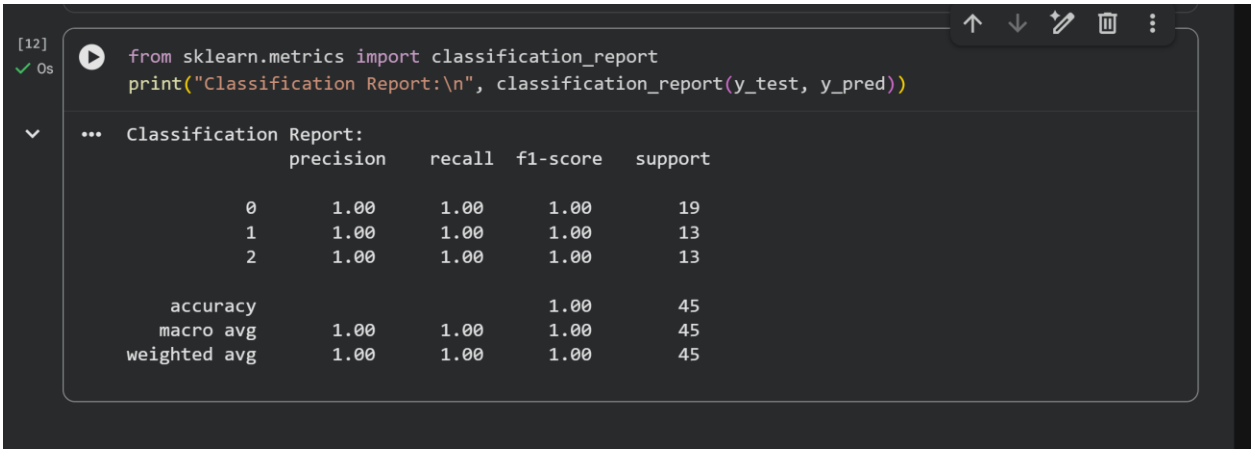
[10]
✓ 0s
from sklearn.metrics import accuracy_score
print("Accuracy Score:", accuracy_score(y_test, y_pred))

Accuracy Score: 1.0
```



```
[11]
✓ 0s
from sklearn.metrics import confusion_matrix
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))

Confusion Matrix:
 [[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```



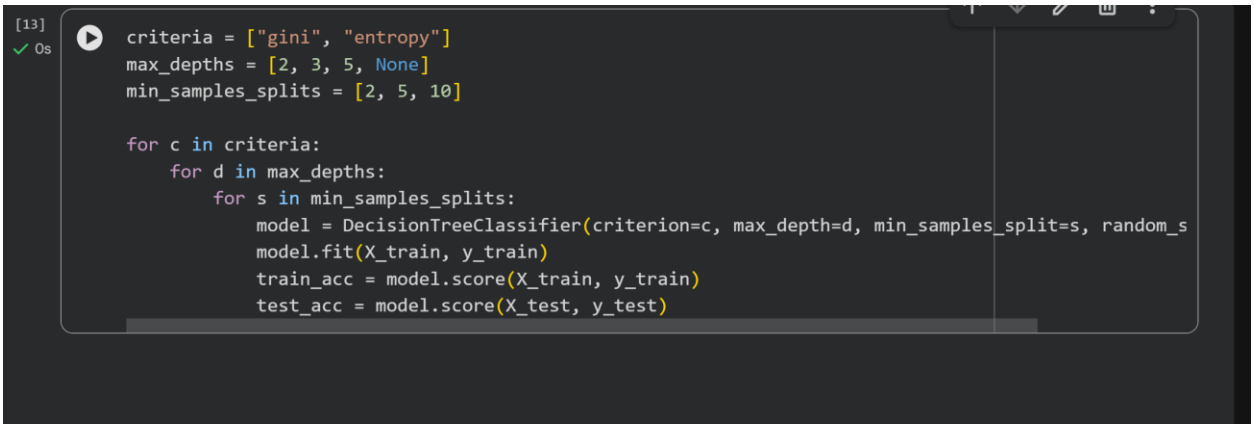
```
[12]
✓ 0s
from sklearn.metrics import classification_report
print("Classification Report:\n", classification_report(y_test, y_pred))

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        19
           1       1.00      1.00      1.00        13
           2       1.00      1.00      1.00        13

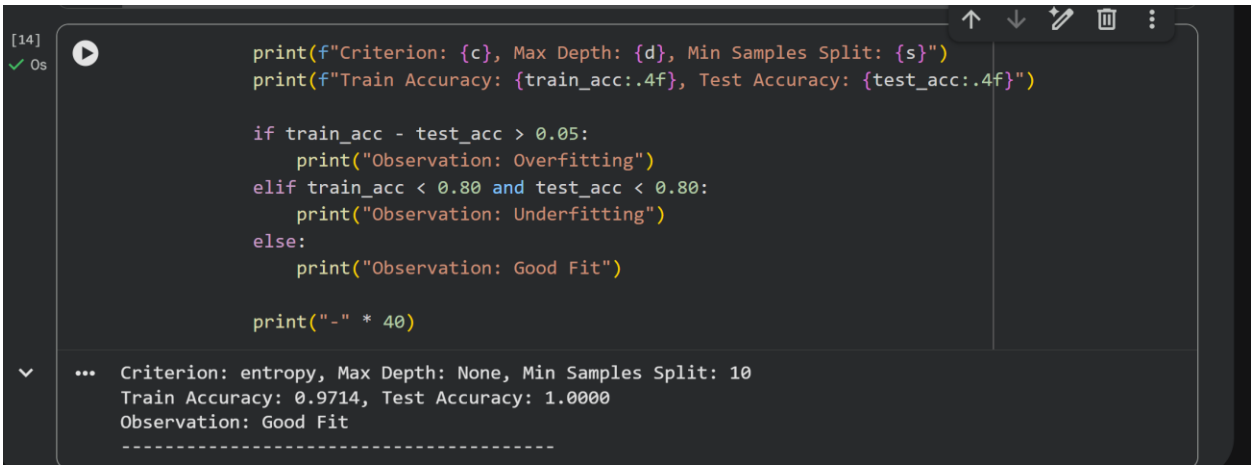
    accuracy                           1.00        45
   macro avg       1.00      1.00      1.00        45
weighted avg       1.00      1.00      1.00        45
```

Task 6:



```
[13]
✓ 0s
criteria = ["gini", "entropy"]
max_depths = [2, 3, 5, None]
min_samples_splits = [2, 5, 10]

for c in criteria:
    for d in max_depths:
        for s in min_samples_splits:
            model = DecisionTreeClassifier(criterion=c, max_depth=d, min_samples_split=s, random_s
            model.fit(X_train, y_train)
            train_acc = model.score(X_train, y_train)
            test_acc = model.score(X_test, y_test)
```



```
[14]
✓ 0s
            print(f"Criterion: {c}, Max Depth: {d}, Min Samples Split: {s}")
            print(f"Train Accuracy: {train_acc:.4f}, Test Accuracy: {test_acc:.4f}")

            if train_acc - test_acc > 0.05:
                print("Observation: Overfitting")
            elif train_acc < 0.80 and test_acc < 0.80:
                print("Observation: Underfitting")
            else:
                print("Observation: Good Fit")

            print("-" * 40)

...
Criterion: entropy, Max Depth: None, Min Samples Split: 10
Train Accuracy: 0.9714, Test Accuracy: 1.0000
Observation: Good Fit
----------------------------------------
```